



Infrequent compost applications increased plant productivity and soil organic carbon in irrigated pasture but not degraded rangeland

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ABSTRACT

Improved agricultural soil management can facilitate the removal of carbon dioxide from the atmosphere to help keep planetary warming at or below 2 °C as outlined in the Paris Agreement. The application of compost to agricultural soils increases soil carbon by directly fertilizing the soil with carbon (no net carbon dioxide removal), and by stimulating plant productivity and plant-derived carbon inputs (net carbon dioxide removal). In semi-arid managed grasslands, it is unclear how compost application may impact plant production and soil carbon stocks. We conducted a multi-year experiment measuring plant and soil carbon and nitrogen responses following additions of compost in two adjacent semi-arid grasslands: an irrigated cool season perennial pasture and a degraded rangeland. Compost application altered plant production and soil carbon and nitrogen dynamics in the irrigated pasture, but not in the degraded rangeland. Aboveground biomass increased approximately 40% under the compost amendment in the irrigated pasture, while belowground biomass only differed between treatments in the first experimental year. Over eight years, soil organic carbon stocks increased 0.6 Mg OC ha⁻¹ yr⁻¹ in the 0–10 cm depth of the irrigated pasture, which was a net increase of 0.3 Mg OC ha⁻¹ yr⁻¹ after accounting for the carbon supplied by the compost. Soil inorganic carbon was highly variable at both experimental sites contributing to about half or more of total soil carbon at depth. Soil inorganic carbon decreased over the 8 year study period through the entire profile (0–50 cm) in the compost amended irrigated pasture ($P = 0.01$) and but not in the other systems, demonstrating the importance of accounting for soil inorganic carbon in carbon stock change estimates under improved soil management practices in semi-arid climates. We conclude that moderate, infrequent applications of high-quality compost to irrigated pastures can promote plant productivity and soil organic carbon, but degraded rangelands may be less responsive to this practice.

1. Introduction

By 2030, global greenhouse gas (GHG) emissions must decline to below current emissions levels if humanity is to meet the goals of the Paris Agreement (Millar et al., 2017). To meet the assumptions outlined in most Intergovernmental Panel on Climate Change (IPCC) mitigation scenarios, humanity must simultaneously reduce anthropogenic GHG emissions while removing large amounts of carbon dioxide (CO₂) from the atmosphere (Hoegh-Guldberg et al., 2018). Carbon dioxide removal strategies via natural climate solutions will be critical to keep planetary warming well below 2 °C (Griscom et al., 2017; Bossio et al., 2020).

Improved management of working lands provides numerous

opportunities for CO₂ removal in the near-term (National Academies of Sciences, 2018), primarily through increased carbon (C) capture and storage in soils (Amelung et al., 2020). Soils contain more C than biomass and atmospheric C reservoirs combined (Batjes, 1996), but historical agricultural cultivation has resulted in an estimated loss of 116 Pg of C from soils (Sanderman et al., 2018). The restoration of C in agricultural soils can be achieved by increasing the inputs of organic C to the soil, and protecting this C from losses, mainly as CO₂, back to the atmosphere.

Grassland and rangeland ecosystems represent half of the land area in the U.S. offering considerable potential to capture and store atmospheric C through improved management (Reeder and Schuman, 2001).

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In particular, the application of compost to managed grasslands may augment the C storage potential of these ecosystems (Ryals and Silver, 2013; Ryals et al., 2014; Gravuer et al., 2019), while also improving soil health (Stott and Moebius-Clune, 2017; Acharya et al., 2019) and economic benefits to land managers (Bowman et al., 2016). Grassland fertilization with organic forms of nitrogen (N) can increase soil organic C (SOC) stocks upwards of $0.54 \text{ Mg C ha}^{-1} \text{ yr}^{-1}$ (Conant et al., 2017). In an annual California grassland, a composted organic amendment (C:N 11) significantly increased net ecosystem C storage over three years by increasing above and belowground net primary production (NPP) (Ryals and Silver, 2013). The increases in SOC exceeded the amount of C input from the amendment itself indicating a net soil C gain through greater annual plant C inputs.

However, more than half of U.S. grassland and rangeland ecosystems occur in water-limited arid and semi-arid environments (Sobecki et al., 2001). Just a fraction of these systems receive irrigation – approximately three percent of all permanent pasture acres in the U.S. (USDA NASS, 2019). There is evidence to suggest that the application of exogenous N to semi-arid grasslands may not stimulate aboveground NPP without concurrent applications of water (Lü et al., 2014). In a long-term study of chronic N deposition at three semi-arid grassland sites in the Colorado Plateau with ambient precipitation, plant communities were not sensitive to N additions indicating resistance of these communities to change (Phillips et al., 2020). Moreover, these communities are also vulnerable to annual invasive species, like *Bromus tectorum*, that can substantially modify plant communities by altering fire regimes and ecosystem C cycling (Bishop et al., 2019). The application of compost to semi-arid grasslands and rangelands requires careful consideration of these challenges because N availability can favor *B. tectorum* relative to native perennial bunch grasses (Vasquez et al., 2008).

For organic amendments like compost to be considered a scalable natural climate solution, more research is required under different environmental conditions to understand the capacity of this management practice to increase soil C storage in the short and medium-term in semi-arid regions. While previous research has shown increased plant cover and soil organic matter levels in the top 8 cm of the soil following biosolid applications to a Colorado semi-arid system (Ippolito et al., 2010), it remains unclear how compost applications may impact soil C cycling at depth. Increases in root biomass and rooting depth stimulated by amendment applications may have consequences for deeper soil C pools. Greater inputs of organic substrates deeper in the soil profile (i.e., root exudates) can stimulate microbial activity and lead to priming of SOC decomposition (Fontaine et al., 2007; Bernal et al., 2016).

Soil inorganic C (SIC), however, can often account for fifty percent or more of total soil C in semi-arid and arid regions, particularly at depth (Plaza et al., 2018). Soil inorganic carbon, usually in the form of calcium carbonate, has longer mean residence times than most SOC and it is governed by complex abiotic and biotic formation and dissolution processes (Monger et al., 2015). It is notoriously difficult to determine whether carbonate formation is a net source or sink of CO_2 from the atmosphere (the source of calcium is an important determinate whether inorganic carbon formation is a sink; Monger et al., 2015), but it can probably be assumed that SIC losses at the farm-scale likely represent a net loss of CO_2 from the system (Sanderman, 2012). Ignoring both SOC and SIC pools at deeper soil depths (> 30 cm) jeopardizes the conclusions about the impacts of agricultural management strategies as natural climate solutions (Tautges et al., 2019). Improved C accounting and consideration of ecosystem C balance under compost will help clarify the feasibility of this practice for adoption in semi-arid regions (Garcia-Franco et al., 2018; National Academies of Sciences, 2018) and increase confidence in incentives and policies for soil C management (Paustian et al., 2016; Amelung et al., 2020).

The objectives of our study were (1) to quantify the impacts of compost applications on ecosystem-level C and N dynamics, and (2) determine if infrequent compost applications are a viable short-term strategy for building SOC in low input semi-arid pasture and

rangeland systems under different types of management. We hypothesized that moderate, infrequent applications of compost would result in a net ecosystem C sink through the stimulation of aboveground biomass and root production leading to greater plant C inputs and subsequent increases in SOC stocks and no net SIC change. We hypothesized that the increases in soil C would be greater than the contributions from the compost, indicating a net capture of atmospheric CO_2 . To test these hypotheses, we conducted a multi-year experiment in northern Colorado on an irrigated pasture (2012–2020) with a history of amendment application and an adjacent degraded rangeland (2018–2020) with no history of compost.

2. Methods

2.1. Study site and general management history

Two experiments were conducted on a working ranch in northern Colorado, USA at an elevation of 1574 m above sea-level ($40^\circ 24' 26.7156'' \text{ N}$, $105^\circ 11' 16.6272'' \text{ W}$). One experiment was on an irrigated pasture/hayland (IP) and a second experiment was on degraded rangeland (DR). Field data were collected for three growing seasons across both experiments from 2018 to 2020, and additional soils data from 2012 to 2020 in IP. The climate in this region is semi-arid characterized by cool winters and warm summers. Precipitation is episodic with most of the precipitation falling during the growing season. The growing season typically begins in early spring (March–April) and ends in the late summer to early autumn months (August–October). The mean annual precipitation (MAP) at the site is 421 mm (30-yr mean), the mean annual temperature (MAT) is 9.5°C , and the mean maximum and minimum temperatures are 18°C and 1.2°C , respectively (30-yr mean, 4 km resolution) (PRISM Climate Group, 2020). Local daily precipitation and temperature data were collected from an on-site weather station installed in August 2018 (Fig. 1). Because the onsite weather station was installed in August 2018, half-way through the growing season, we also analyzed weather data for 2018–2020 from PRISM Climate Group (2020) at a 4 km resolution to compare relative precipitation amounts across the three study years.

From the mid-1970 s to late 1990s the entire site (IP, DR) was continuously grazed by horses until cattle were introduced around the year 2000. Continuous grazing led to bare areas and invasion by annual weeds – annual ragweed (*Ambrosia artemisiifolia* L.) and kochia (*Kochia scoparia* L. Schrad.) – in sections of the overall site. Rotational grazing was introduced in 2005 to combat these issues. A light-weight tine harrow was dragged across the site every three to four years for the purposes of spreading manure in the spring before the growing season, tine depth was approximately 3 cm and did not result in a large soil or vegetation disturbance. The entire site is also part of a corridor of seasonal grazing by Rocky Mountain elk (*Cervus canadensis nelsoni*) in the early spring and late autumn.

The IP, which was in a lower lying area of the site, was primarily managed for hay during the field experiment. The IP vegetation consisted of perennial alfalfa (*Medicago sativa* L.) and introduced cool season grasses (e.g., meadow brome, *Bromus commutatus* Shrad., smooth brome, *B. inermis* L., and orchard grass, *Dactylis glomerata* L.) that were harvested for hay in the summer and sometimes grazed in the winter months. Irrigation of IP began in the 1970 s with side roll irrigation and a center pivot was installed in 2007. Higher elevation areas at the site were maintained as DR that were grazed but never seeded or harvested for hay. The DR plant community was comprised of a mixture of exotic and native cool and warm-season grasses, legumes, forbs, and shrubs (e.g., intermediate wheatgrass, *Thinopyrum intermedium* L., alfalfa, *M. sativa*, least pepperwort, *Lepidium virginicum* L., and fringed sage, *Artemisia frigida* Willd.). Exotic invasive annuals like cheatgrass, *Bromus tectorum* L., and rye brome, *Bromus secalinus* L., were also found within these pastures. Beef cattle were rotated through these pastures year-round, and grazing varied in timing and frequency.

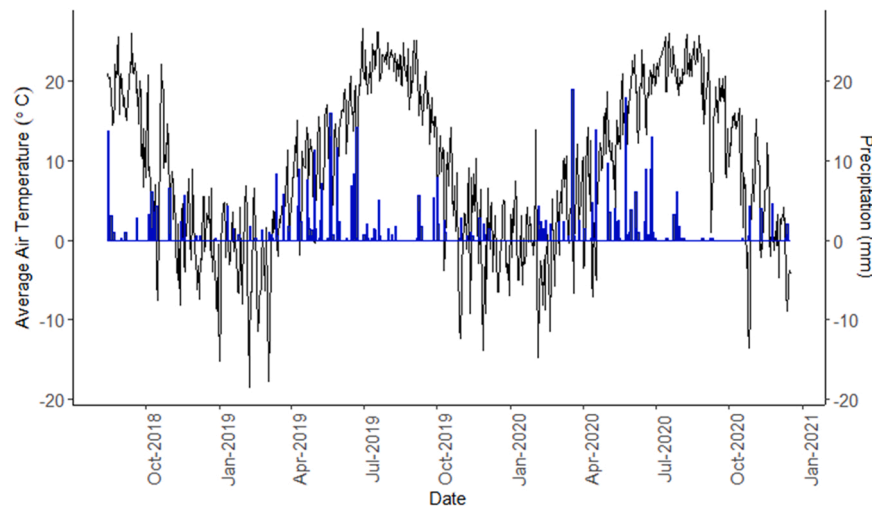


Fig. 1. The daily average air temperature (°C) in black and b) daily precipitation (mm) in blue from August 2018 to December 2020 at the experimental site collected from an onsite weather station.

The soils in the study area are mesic Aridic Arguistolls with 0–3.5% (IP) and 0–5% (DR) slopes (Soil Survey Staff, 2017; USGS, 2021). The primary soil in IP is a Nunn clay loam (fine, smectitic, mesic Aridic Arguistoll) and the DR Connerton-Sylvan Dale complex (fine-loamy, mixed, superactive, mesic Torriorthentic Haplustolls-Pachic Arguistolls) and Kirtley loam (fine-loamy, mixed, superactive, mesic Aridic Arguistoll). The parent material of these soils is alluvium derived from sandstone and shale. A complete description of baseline soil characteristics and textures for each experiment can be found in Table 1.

2.1.1. Experiment design – Irrigated pasture (IP)

The IP experiment was established in 2012 with two levels of treatment – compost addition and unamended control with eight replicated blocks (Fig. 2). The plots measured 10 × 10 m. Commercially available composted livestock manure (Humalfa; Iliff, Colorado, USA) was applied twice; once in March 2012 and again in March 2018. For this study we focused on data collected in the years 2018–2020 after the second application, but differences in soil C and total N stocks were evaluated from 2012 to 2020. At both applications, the compost was applied at a rate of 11.2 Mg ha⁻¹ (C:N:P:K ratio 8:1:1:2) by compost spreader, which was equivalent to 123 kg N ha⁻¹, 974 kg organic C ha⁻¹, 171 kg P ha⁻¹, and 212 kg K ha⁻¹. The unamended control plots were carefully marked and given a five-meter buffer to avoid contamination by wind. A single hay cutting occurred in June or July each experimental year, and the pasture was occasionally grazed by beef cattle in the winter months (periodic moderate, continuous grazing). A description of management and sampling dates can be found in Table 2.

2.1.2. Experiment design – degraded rangeland (DR)

The DR experiment was established in March 2018 as a randomized complete block design and there were two levels of treatment: compost

and an unamended control with six replicated blocks. The compost treatment was the same compost material applied in IP, but it was applied at a lower rate in DR to better reflect potential investment a rancher might make on degraded rangeland with lower potential productivity. In DR, the compost was applied at a rate of 6.7 Mg ha⁻¹ (C:N:P:K ratio 8:1:1:2) by compost spreader on March 25, 2018, which was equivalent to 74 kg of total N ha⁻¹, 583 kg organic C ha⁻¹, 103 kg P ha⁻¹, and 127 kg K ha⁻¹. More management details are available in Table 2.

Grazing by beef cattle in the DR was managed by different cattle leases over the three growing seasons. The stocking rate ranged from 15.8 to 23.8 head ha⁻¹ and followed rotational grazing management. In 2018 and 2019, the beef cattle were rotated once through two blocks of DR (approximately 1.26 ha) beginning in September for a period of 10–14 days until all plots were grazed. During the final growing season, beef stockers rotated through only four of the replicate blocks of DR from late April to early May for upwards of seven days (Table 2).

2.2. Above biomass

Aboveground biomass estimates were made during peak standing biomass in IP and DR. In IP, these samples were collected prior to haying in June or July of each year, and in DR they were collected in late July or early August prior to cattle grazing. We clipped biomass from two 0.25 m² quadrats in each plot about 2 centimeters above the soil surface to avoid soil or manure contamination. Biomass samples were dried at 55 °C immediately after collection for at least 48 h, and then weighed for mass on an oven-dried basis. All samples were ground with a ball mill and analyzed for C and N concentration on a LECO Tru-Spec CN analyzer (Leco Corporation; St. Joseph, Minnesota, USA).

Table 1

Baseline soil properties (mean ± SE) of each field experiment measured in 2012 for the irrigated pasture (IP) and in 2018 for the degraded rangeland (DR). Soils were sampled to bedrock, which was around 50 cm in IP and 80 cm in DR. The reported soil carbon (C) and nitrogen (N) stocks were quantified using equivalent soil mass.

Field experiment	Depth (cm)	Clay (%)	Silt (%)	Sand (%)	pH	Bulk Density (g cm ⁻³)	Soil organic C (Mg C ha ⁻¹)	Soil inorganic C (Mg C ha ⁻¹)	Soil N (Mg N ha ⁻¹)
IP	0–10	34.6	19.9	45.4	7.8	1.01 ± 0.01	18.9 ± 0.8	10.5 ± 1.9	2.3 ± 0.05
	10–20	34.3	19.3	46.5	8.0	1.14 ± 0.01	12.3 ± 0.5	11.1 ± 1.9	1.8 ± 0.04
	20–50	35.0	24.2	40.8	8.0	1.09 ± 0.02	14.5 ± 1.7	57.2 ± 7.6	3.2 ± 0.11
DR	0–10	44.7	23.0	32.3	8.1	1.33 ± 0.05	19.5 ± 1.0	7.0 ± 0.8	2.1 ± 0.1
	10–20	45.2	21.2	33.7	8.3	1.40 ± 0.04	13.7 ± 1.0	10.8 ± 0.8	1.7 ± 0.1
	20–50	34.0	20.7	45.3	8.1	1.39 ± 0.03	25.6 ± 2.3	40.8 ± 4.0	2.9 ± 0.1
	50–80	32.3	25.3	42.3	8.2	1.42 ± 0.03	19.0 ± 1.2	53.0 ± 2.9	3.7 ± 0.1

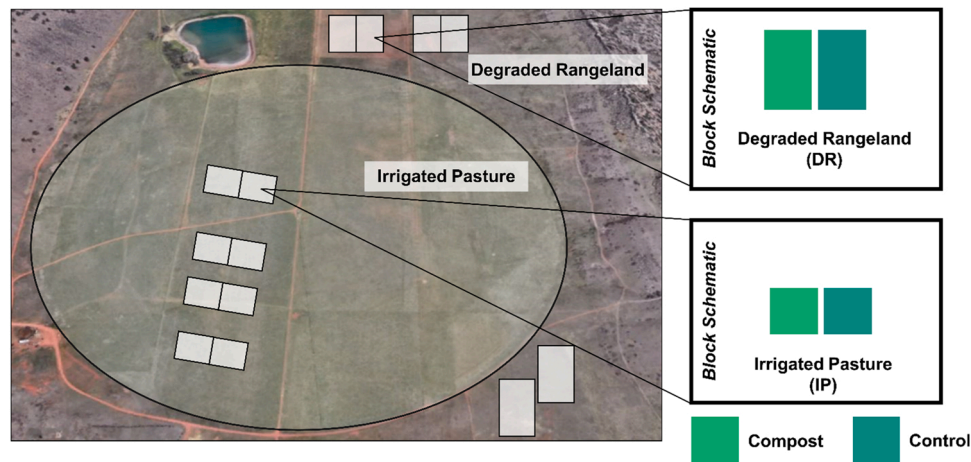


Fig. 2. Replicated experimental plot design for the irrigated pasture (IP) with 8 blocks and the degraded rangeland (DR) with 6 blocks.

Table 2

Management history and operations timeline for field experiments. IP: irrigated pasture; DR: degraded rangeland.

	IP				DR		
	2012	2018	2019	2020	2018	2019	2020
<i>Management</i>							
Compost	11.2	11.2	—	—	6.7	—	—
Amount applied (Mg ha ⁻¹)	123	123	—	—	74	—	—
N rate (kg N ha ⁻¹)	March	March 23	—	—	March 26	—	—
Date applied	27						
Annual Irrigation Amount (mm ha ⁻¹ yr ⁻¹)	—	408	292	775	—	—	—
Hay cutting	—	July 10	July 21	June 18	—	—	—
Grazing	—	Continuous (winter) ^a	Continuous (winter) ^a	None	Rotational (September–December)	Rotational (September–January)	Rotational (April, May) ^b
<i>Sampling</i>							
Baseline soil samples	March 26	—	—	—	March 13, April 26 ^c	—	—
Root biomass	March 26	March 12, August 06	August 01	March, July 01	March, September–November	—	September 17
Soil measurements	—	March 12, August 06	August 01	March 25, July 01	March, September–November	—	September 17
Aboveground biomass	—	June 12	July 02	June 17	August 06	July 29	July 20
Soil CO ₂ Flux	—	—	—	—	April 12 – October 19	April 08 – October 09	April 15 – October 01

^a During the winter when cattle grazing occurred, there was minor feed supplementation with imported hay.

^b As a result of grazing management decisions, only four of six replicate plots were grazed in 2020.

^c Bulk density samples.

2.3. Grazed biomass removal in DR

To estimate biomass removal by grazers, we used the moveable cage method to determine total grazed biomass as described by McNaughton et al. (1996). In early April of each experimental year, four metal cages (1.1 m diameter, 0.6 m high) were randomly installed in each DR plot. After each grazing event, a 0.25 m² quadrat was placed inside the cage to determine the amount of pre-grazed above-biomass, and another quadrat was placed outside and immediately adjacent to the cage to estimate post-grazed above-biomass, which was determined as described above. The difference between pre- and post-grazed samples after multiplying oven dried mass by the C concentration (determined as described above) represented the estimated biomass removal (Mg C ha⁻¹) by the cattle. Our estimates of grazed biomass C removal are likely high because we did not estimate C returned to the soil through cattle excrement.

2.4. Belowground biomass, soil carbon, nitrogen, and physical properties

We quantified belowground biomass, field bulk density, and all soil

physiochemical properties from the same cores (4 cm inside diameter) collected with a Giddings hydraulic soil probe (Giddings Machine Company; Windsor, Colorado, USA). Three cores were collected per plot at three depth intervals (0–10, 10–20, and 20–50 cm) in IP in 2012 prior to first organic amendment application. In 2018–2020, we collected five (0–10, 10–20 cm) and three soil cores (20–50 cm) from IP. In DR, we collected five soil cores (4 cm inside diameter) per depth interval (0–10, 10–20, 20–50, 50–80 cm) per plot due to the larger plot size. Soil cores were bulked in the field and stored at 4 °C until processing. Immediately after returning from the experimental site, we recorded sample mass and removed a 12 g subsample for gravimetric water content analysis. These subsamples were placed in an oven for at least 48 h at 105 °C. Bulk density estimates were mass corrected for coarse particles > 2 mm diameter following the recommended approach for dryland soils in Throop et al. (2012).

Standing root biomass estimates were made to 20 cm in both IP and DR using the soil core method (Milchunas, 2009). Previous analyses of the proportion of roots from 0 to 50 cm in IP indicated that less than 20% of roots occurred below a 20 cm depth. Thus, we only collected roots to 20 cm depth during the growing season for both experiments.

For IP, root biomass was collected annually over the duration of the experiment. In DR, root biomass was collected in 2018 during the first growing season two weeks after grazing by beef cattle (September – December; 0–20 cm) and again in September 2020 (0–20 cm).

All roots were extracted by hand from the bulked soil samples. We partitioned roots into two different fractions. The first fraction (coarse roots) was obtained by collecting all visible roots from the sample before passing through an 8 mm sieve. For the second fraction, a 150 g air dried soil subsample was used to collect fine roots before passing the soil subsample through a 2 mm sieve. All soil was carefully removed from the roots before drying at 55 °C to obtain oven-dried mass. The sum of coarse and fine root fractions represented total belowground standing biomass per depth interval. Consistency in the efficiency of root extraction was difficult to maintain across years, thus, we primarily focus on differences between treatments within year in our analyses. To estimate belowground standing biomass, we multiplied the oven-dried mass of the root sample by a 45% C concentration value (Ma et al., 2018).

We quantified total soil C and N concentrations on 2 mm sieved soils that were ground and dried at 55 °C using a LECO Tru-Spec CN analyzer (Leco Corporation; St. Joseph, Minnesota, USA). We quantified SOC as the difference between total soil C and SIC. We measured SIC concentration on all soil samples (Table 1) using the pressure-calimeter method (Sherrod et al., 2002). Due to the high variability and large concentrations of SIC, particularly at deeper soil depths (Table 1), we also measured SOC for a subset of soil samples by dry combustion on a CN analyzer, after removing SIC through acid fumigation (Harris et al., 2001). The results from both methods were highly correlated ($R^2 = 0.93$, Slope = 0.97), but three of 48 samples in IP fell well outside the 1:1 correlation when comparing methods. We elected to remove these outlier samples from our analyses of soil C stocks. Soil C and N concentrations were converted to C and N stocks using the equivalent soil mass (ESM) method described in Lee et al. (2009). Soil texture and soil pH were analyzed by Ward Laboratories, Inc. (Lincoln, Nebraska, USA) by hydrometer (Gee and Bauder, 1986) and a 1:1 slurry of soil and deionized water, respectively.

2.5. Soil respiration

Because we expected changes in SOC to be difficult to detect during the three-year experimental period with the low compost application rates in DR, we also measured soil respiration in DR plots. Soil CO₂ flux can be a more responsive measurement to treatment effects on soil C dynamics. The measurements were collected twice weekly during the first growing season (2018) beginning in April unless a rain event prevented data collection. We collected measurements at this frequency to better capture the anticipated increase in CO₂ fluxes immediately after amendment application. The sampling frequency was reduced to once weekly in second growing season (2019) and bi-monthly in the third growing season (2020). Briefly, at least 24 h prior to the first measurement date, PVC collars were installed 5 cm into the soil surface and allowed to equilibrate. One PVC collar was installed per plot for a total of 6 observations per treatment. Soil CO₂ flux was measured with a LI-8100 infrared gas analyzer and survey chamber system (~ 2-minute flux intervals [LI-COR Biosciences, Lincoln, Nebraska, USA]). Soils within the collars were kept free of plant shoots to provide an estimate of soil respiration only. To scale to an approximation on seasonal soil CO₂ flux, we applied a linear interpolation (Bremer et al., 1998; Ryals and Silver, 2013) method between sampling time points and summed the estimated daily fluxes over the entire sampling period (Table 2).

2.6. Statistical analyses

We analyzed all data using linear mixed models using the *lme4* package (Bates et al., 2014) in R version 3.6.3 (R Core Team, 2013). The IP and DR experiments were analyzed separately. To assess change over

time we used repeated measure multivariate analysis of variance (MANOVA) where there were multiple years of observations. Treatment and year were fixed effects and block and plot were random effects. When we had observations at a single time point, we applied one-way analysis of variance (ANOVA) to test for statistically significant treatment effects; block and plot were included as random effects. Degrees of freedom were corrected according to the method recommended by Kenward and Roger (1997). We included baseline 2012 root biomass, soil C, soil N, and bulk density responses at the plot level in IP as covariates due to initial differences between treatment and control plots for these variables. Linear interpolation of daily soil CO₂ flux rates to seasonal cumulative soil CO₂ flux yr⁻¹ was performed using the *PKNCA* package in R (Buckeridge et al., 2015). Soil C and N concentrations were converted to ESM using the *SimpleESM* R script (Ferchaud, 2020). Outliers were removed in rare instances when they exceeded the 25 or 75% quartile ranges impacting less than 6% of data points for any given variable. Significant differences between treatments and treatment by year were assessed with pairwise comparisons using the *emmeans* package (Lenth, 2020) in R to estimate least-squares means. Data were reported as least squares means followed by standard error, and we determined statistical significance at $P \leq 0.05$ unless otherwise noted.

3. Results

3.1. Annual weather and irrigation

Using the onsite weather station, the annual precipitation was 235 mm in 2019, and in 2020 it was 196 mm. Annual precipitation was highest in 2019, according to the PRISM weather data, with about 10% and almost 20% lower precipitation amounts in 2018 and 2020, respectively. The annual temperature was lower in 2019 relative to 2018 and 2020 by about 1.5 °C. In IP, annual irrigation rates were 47% (367 mm ha⁻¹ yr⁻¹) and 62% (483 mm ha⁻¹ yr⁻¹) lower in 2018 and 2019, respectively, relative to 2020 (Table 2). Irrigation in 2019 was lower than 2018 and 2020 due to equipment malfunction.

3.2. Aboveground biomass carbon and nitrogen responses

The compost treatment increased aboveground biomass production during the experimental period in IP; in 2018, 2019, and 2020 the compost treatment yielded, respectively, 4.2 ± 0.3 , 2.9 ± 0.3 , and 4.5 ± 0.3 Mg ha⁻¹ yr⁻¹ while the control yielded, respectively, 3.0 ± 0.3 , 2.5 ± 0.3 , and 3.3 ± 0.3 Mg ha⁻¹ yr⁻¹. In 2018 and 2020, aboveground biomass C and N were 37% and 42% greater, respectively, where compost had been applied than the unamended control ($P < 0.01$; Fig. 3a,c). While this trend persisted in 2019, overall biomass C was 30% lower in 2019 across all treatments than the other years likely due to poor irrigation and the treatment means did not differ ($P > 0.05$). There were no differences in aboveground biomass C:N between treatments; biomass C:N across both treatments ranged from 25.9 to 32.1 (Table A.1).

In DR, the compost treatment resulted in marginally higher mean aboveground biomass, and biomass C and N during the first (1.6 ± 0.4 vs. 1.1 ± 0.4 Mg ha⁻¹ yr⁻¹) and second (2.2 ± 0.4 vs. 2.0 ± 0.4 Mg ha⁻¹ yr⁻¹) experimental years relative to the control, but there were only detectable differences for aboveground N in 2018 ($P < 0.05$). In the third experimental year the control had higher mean aboveground biomass (1.7 ± 0.4 vs. 1.5 ± 0.4 Mg ha⁻¹ yr⁻¹) and biomass C and N (Fig. 3c,d) relative to the compost, but there no differences between treatments. We also did not observe any differences between treatments for aboveground biomass C:N (Table A.1). The amount of biomass removed annually by grazers in DR did not differ between treatments or years. However, grazed biomass tended to be lower in 2020, which was likely attributable to shorter grazing duration times (Table A.2).

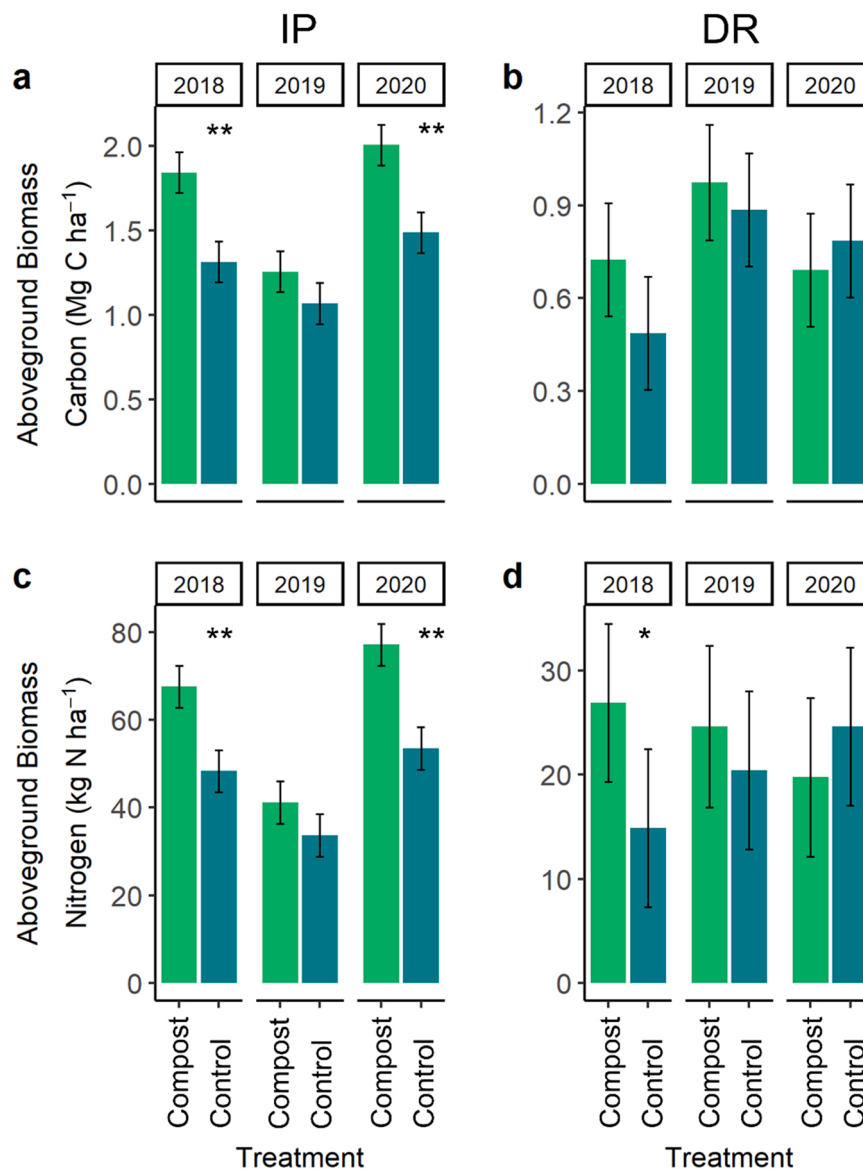


Fig. 3. Annual least-squares means of aboveground biomass carbon (Mg C ha⁻¹) responses for a) irrigated pasture (IP) and b) degraded rangeland (DR) and aboveground biomass nitrogen (kg N ha⁻¹) for c) IP and d) DR during a three-year experiment near Loveland, CO. Significant p-values for treatments within year; $P \leq 0.05$ (*), $P \leq 0.01$ (**). Error bars represent standard error; IP ($n = 8$) and DR ($n = 6$).

3.3. Root standing biomass

Standing root biomass C varied by year in IP. Estimates of root biomass collected during peak standing aboveground biomass differed between treatments in IP in the first study year (Fig. 4a, Table A.1). The compost treatment had 23% greater standing root biomass C from 0 to 20 cm in 2018 (Table A.1). However, there were no other differences between treatments from 0 to 20 cm in 2019 or 2020.

There were no differences in root biomass between C treatments when estimated from 0 to 20 cm in DR (Fig. 4b, Table A.1). Estimates of root biomass C also varied annually, particularly in the 0–10 and 10–20 cm depths, which were sampled twice over the three-year experiment (Fig. 4b). Root biomass C was notably lower in 2020 relative to other years in both IP and DR.

3.4. Bulk density and soil carbon and nitrogen stocks

Soil bulk density did not differ between treatments in the 0–10 cm or 10–20 cm depths over the three-year experiment in IP (Table A.3).

However, soil bulk density was 9% lower in 2020 in 20–50 cm depth in the unamended control relative to the compost treatment. All depth intervals showed decreases in bulk density from 2018 to 2020; soil bulk density was about 12% lower across all depth intervals in 2020 relative to 2018 (Table A.3). Similarly, in DR, there were no differences between treatments in bulk density at any depth interval.

Soil organic C and N stocks were greater in the compost treatment relative to the control in IP but not DR. Soil C stock estimates were variable across all treatments, particularly for SIC, while total soil N showed less variability. We quantified equivalent soil mass SOC, SIC, and total N stocks in the topsoil (0–10 cm) and at deeper soil intervals to bedrock (50 cm and 80 cm in IP and DR, respectively) and analyzed stocks for treatment effects at each sampling point. In DR, there were no differences between treatments in any soil C or total N stocks from 0 to 10 cm or 10–80 cm in 2020 (Table 3). There were no differences between treatments for soil C:N at any depth interval, but across treatments soil C:N decreased with depth (Table A.4).

In IP, SOC stocks from 0 to 10 cm were 16% higher in the compost treatment relative to the control in 2020 ($P < 0.01$), and only 8% higher

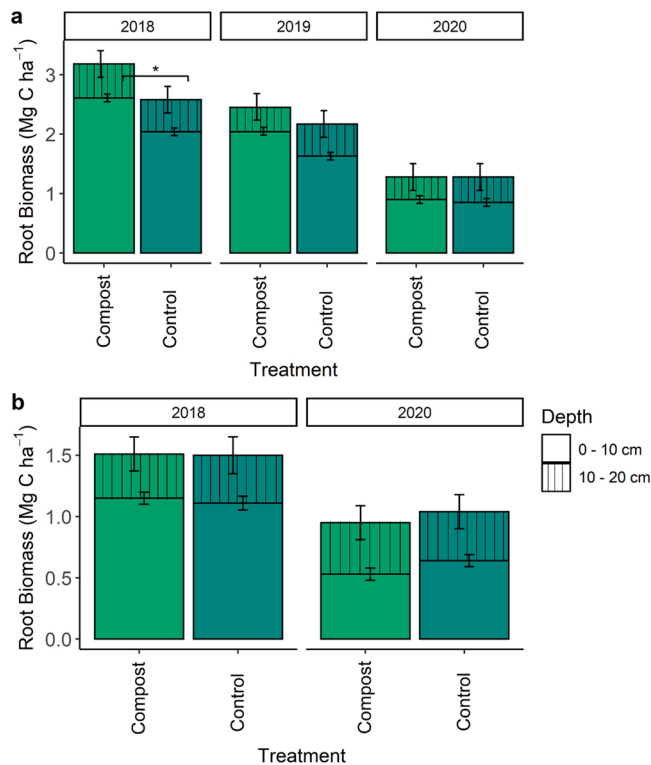


Fig. 4. Annual least-squares means of standing root biomass carbon (Mg C ha⁻¹) responses for a) irrigated pasture (IP) and b) degraded rangeland (DR) during a three-year experiment near Loveland, CO. Significant p-values for treatments within year; $P > 0.05$ (non-significant; NS), $P \leq 0.05$ (*). Error bars represent standard error; IP (n = 8) and DR (n = 6).

than the control in 2018 prior to the second compost application. Soil inorganic C stocks were 42% higher in the unamended control in 2018 from 0 to 10 cm relative to the compost treatment ($P < 0.05$) but did not differ by treatment in 2020. Total N stocks from the 0–10 cm depth in 2018 and 2020 were 14% and 23% higher in the compost treatment,

respectively, than the unamended control in IP ($P < 0.05$ and $P < 0.01$, respectively).

From 10–50 cm in IP, we only observed differences in total soil N, but not in C stocks between treatments. Total N stocks from 10 to 50 cm in the compost treatment were 9% higher than the control in 2020 (Table 3, $P < 0.05$). When aggregated across the entire soil profile (0–50 cm; Table A.4), SOC stocks were 10% higher under the compost treatment in 2020, but there were no differences in SIC. We also observed differences between treatments throughout the entire soil profile for total soil N in both 2018 and 2020. The strong response of SOC and total soil N stocks to the organic N treatment were concentrated from 0 to 10 cm, with only a couple of differences below 10 cm for C and N stocks (Table A.4).

3.5. Changes in soil carbon and nitrogen stocks

To estimate change in soil C and N stocks over time, we compared baseline measurements collected in 2012 in IP and 2018 in DR to our final year of measurements in 2020. We examined the differences in stocks between treatments and whether changes in stocks were different from baseline measurements (Fig. 5, Table A.5). In DR, there were no changes in soil C or N stocks over the short experimental duration (2018–2020) or differences between treatments (Fig. 5b,d, Table A.5).

Between 2012 and 2020, we measured significant increases in SOC stocks in IP. Soil organic C stocks increased by 4.7 Mg OC ha⁻¹ more over eight years in the compost treatment from 0 to 10 cm than in the unamended control (Fig. 5a). This increase exceeds the total amount of C added to the system by the compost over two applications (1.95 Mg OC ha⁻¹ total). On an annual basis, the compost resulted in an increase of 0.6 Mg OC ha⁻¹ yr⁻¹ more than the control. After subtracting the added compost C from the difference between total change in SOC under the organic amendment versus the control, SOC stocks increased by 2.8 Mg C ha⁻¹ or 0.3 Mg C ha⁻¹ yr⁻¹ from 0 to 10 cm in the compost treatment. However, when accounting for SOC below 10 cm, we did not observe any differences between treatments, but both treatments gained SOC over the period. While there were no significant differences in SIC stocks between treatments at individual sampling points, the compost treatment lost SIC from 10 cm to 50 cm over the eight years since initial application (Fig. 5c). These losses were driven by decreases in SIC at

Table 3

Annual least-squares means $\pm$ standard error (SE) of soil equivalent soil mass (ESM) carbon and nitrogen stocks for control and treatment plots from two field experiments. IP: irrigated pasture (n = 8); DR: degraded rangeland (n = 6). Significant p-values for treatments within year indicated by different asterisks; $P > 0.05$ (non-significant; NS), $P \leq 0.05$ (*), and $P \leq 0.01$ (**).

Variable	Depth (cm)	Year	IP ^a		P	DR ^b		P
			Compost	Control		Compost	Control	
Soil organic carbon (Mg OC ha ⁻¹)	0–10	2018	22.4 $\pm$ 0.8	20.7 $\pm$ 0.8	*	19.5 $\pm$ 1.0	19.5 $\pm$ 1.0	NS
		2020	26.3 $\pm$ 0.8	22.6 $\pm$ 0.8	**	22.2 $\pm$ 1.3	20.9 $\pm$ 1.3	
	10–D ^c	2018	41.3 $\pm$ 2.6	46.8 $\pm$ 2.3	NS	58.3 $\pm$ 3.5	58.3 $\pm$ 3.5	NS
		2020	41.2 $\pm$ 2.6	43.7 $\pm$ 2.3	NS	58.7 $\pm$ 5.5	52.4 $\pm$ 4.8	
	0–D	2018	68.8 $\pm$ 2.2	67.5 $\pm$ 2.0	NS	77.8 $\pm$ 4.3	77.8 $\pm$ 4.3	NS
		2020	72.9 $\pm$ 2.2	66.1 $\pm$ 2.0	*	81.2 $\pm$ 6.6	73.4 $\pm$ 5.8	
Soil inorganic carbon (Mg IC ha ⁻¹)	0–10	2018	8.1 $\pm$ 0.9	11.5 $\pm$ 0.9	*	7.0 $\pm$ 0.8	7.0 $\pm$ 0.8	NS
		2020	9.0 $\pm$ 0.9	10.0 $\pm$ 0.9	NS	4.4 $\pm$ 1.9	8.0 $\pm$ 1.9	
	10–D	2018	46.4 $\pm$ 4.6	53.1 $\pm$ 4.4	NS	105.0 $\pm$ 6.8	105.0 $\pm$ 6.8	NS
		2020	40.4 $\pm$ 4.6	51.2 $\pm$ 4.4	NS	90.0 $\pm$ 19.4	120.0 $\pm$ 17.3	
	0–D	2018	55.7 $\pm$ 4.4	61.0 $\pm$ 3.8	NS	112 $\pm$ 6.9	112 $\pm$ 6.9	NS
		2020	48.9 $\pm$ 4.4	59.2 $\pm$ 3.8	NS	95.0 $\pm$ 20.8	128.0 $\pm$ 18.6	
Total soil nitrogen (Mg N ha ⁻¹)	0–10	2018	2.4 $\pm$ 0.1	2.1 $\pm$ 0.1	*	2.1 $\pm$ 0.1	2.1 $\pm$ 0.1	NS
		2020	2.7 $\pm$ 0.1	2.2 $\pm$ 0.1	**	2.2 $\pm$ 0.1	2.1 $\pm$ 0.1	
	10–D	2018	4.9 $\pm$ 0.1	4.9 $\pm$ 0.1	NS	6.9 $\pm$ 0.2	6.9 $\pm$ 0.2	NS
		2020	4.9 $\pm$ 0.1	4.5 $\pm$ 0.1	*	6.4 $\pm$ 0.4	5.9 $\pm$ 0.4	
	0–D	2018	7.3 $\pm$ 0.2	7.0 $\pm$ 0.2	*	9.0 $\pm$ 0.3	9.0 $\pm$ 0.3	NS
		2020	7.6 $\pm$ 0.2	6.8 $\pm$ 0.2	**	8.6 $\pm$ 0.6	8.0 $\pm$ 0.6	

^a Values from 2018 represent stocks measured over the period of 2012–2018, and values from 2020 represent stocks measured over the period of 2018–2020. These values include 2012 estimates as a covariate due to initial treatment differences at the plot level.

^b Values from 2018 represent baseline stock measurements at the block level, and values from 2020 represent stocks measured over the period of 2018–2020.

^c Full profile depth (D) was 50 cm for IP and 80 cm for DR.

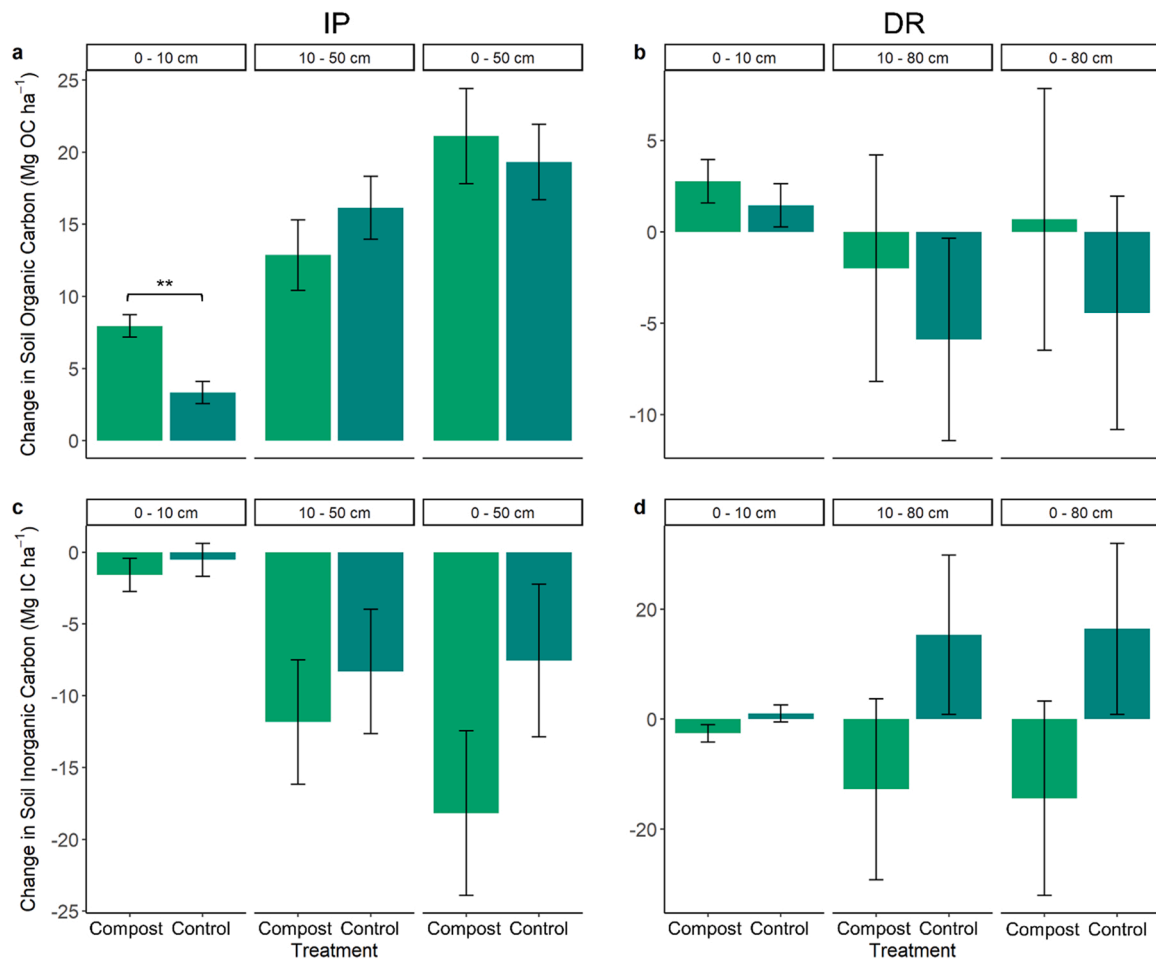


Fig. 5. The mean a) daily soil respiration (R_s) ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) and b) least squares means of cumulative soil R_s (Mg C ha^{-1}) in the degraded rangeland during a three-year experiment near Loveland, CO. Significant p-values for treatments within year; $P > 0.05$ (non-significant; NS), $P \leq 0.05$ (*), and $P \leq 0.01$ (**). Error bars represent standard error; $n = 6$.

depth (20–50 cm).

Total soil N stocks in IP were less variable and demonstrated consistent differences in stock changes between treatments in the top and full soil profile (Table A.5). In the topsoil, total N had a larger increase in the compost treatment relative to the unamended control. From 10–50 cm, soil N in 2020 did not differ from baseline measurements.

3.6. Soil carbon respiration

There were no differences in seasonal cumulative soil respiration between treatments in DR (Fig. 6; Table A.6). However, soil respiration averaged across treatments was highest during the first experimental year when the compost was applied, and lowest in 2020. On average, cumulative soil CO_2 flux was 14% and 45% higher than 2019 and 2020, respectively, when averaged across treatments. Daily CO_2 flux rates were highest earlier in the growing season (April–July) and then steadily declined during late summer and early fall (Fig. 6a).

4. Discussion

Compost applications altered plant productivity and soil C and N dynamics in an irrigated pasture (IP), but not in a neighboring degraded rangeland (DR). The increased above and belowground plant C inputs in IP contributed to increased SOC in the top 10 cm and the full soil profile. Similarly, total N stocks were consistently higher under the compost in the top (0–10 cm) and full soil profile (0–50 cm). However, net changes

in SOC stocks were only greater in the topsoil (0–10 cm) after eight years, and they did not differ by treatments over this time in the deeper soil profile (10–50 cm) or when aggregated across the full profile (0–50 cm). Unlike soil C, the variability in total N stocks did not increase with depth and we were able to detect persistent treatment differences in soil N stocks over time for the full soil profile. But, we did not detect any changes in soil C:N over the experimental period at any depth reflecting the parallel increases in soil C and N under the amendment.

The increase in total N stocks exceeded the N supplied by the compost in IP. A total of $0.25 \text{ Mg N ha}^{-1}$ were added (Table 2) to the soil from the compost applications. During the first application period, 2012–2018, there was a 0.3 Mg N ha^{-1} difference between the compost treatment and the control from the 0–10 cm depth, and from 2018 to 2020 this difference was even greater at 0.5 Mg N ha^{-1} . Across the whole soil profile there was a 0.3 Mg N ha^{-1} difference in 2018 and a 0.8 Mg N ha^{-1} difference between treatments in 2018 and 2020, respectively. This strongly suggests that total soil N stocks increased beyond the N supplied from the compost. In these same plots, we observed an increase in legumes, particularly alfalfa, over the experimental period under the compost (data not shown), which are often limited by other soil macronutrients provided by compost (e.g., phosphorus). The greater prevalence of alfalfa coupled with the increased plant biomass production potentially had higher biological N fixation (BNF) than the control plots, which may explain the total soil N increase beyond what was supplied by the compost. Because biological N fixation demands high amounts of phosphorus (Mitran et al., 2018) it is possible that the phosphorus supplied by the organic amendment alleviated a phosphorus limitation

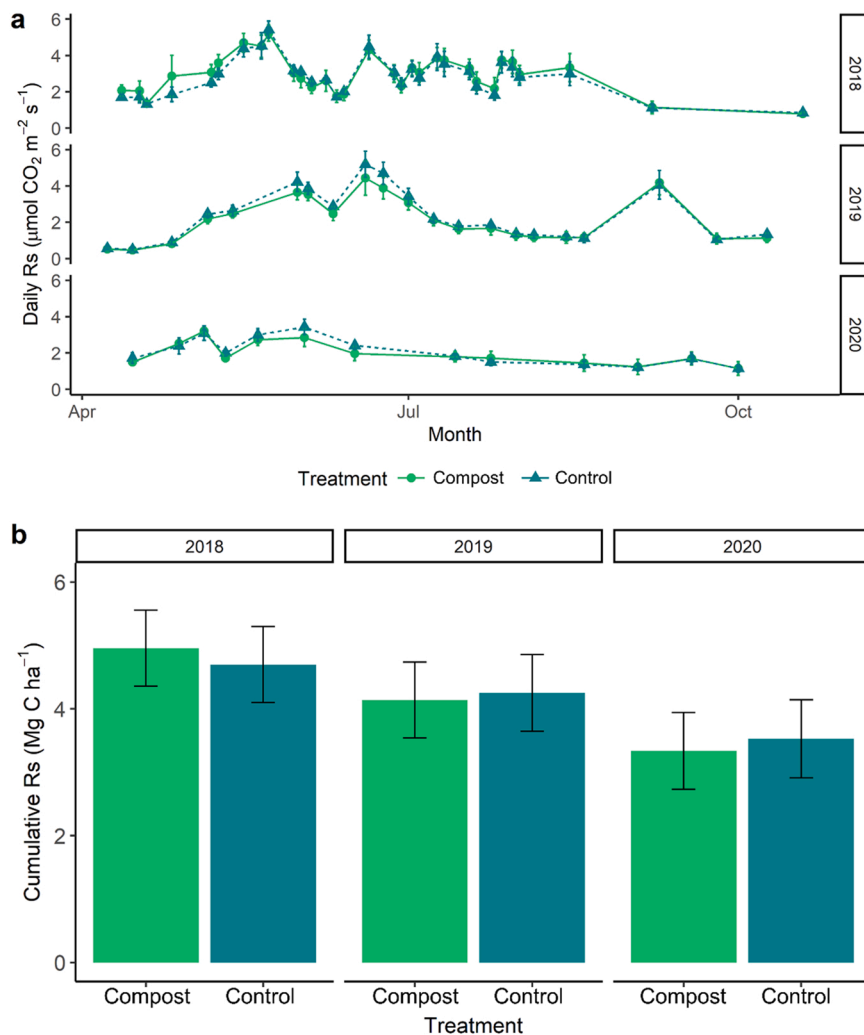


Fig. 6. Annual least-squares means $\pm$ standard error (SE) of soil organic carbon stock changes for a) irrigated pasture (IP) and b) degraded rangeland (DR) and soil inorganic carbon stock changes for c) IP and d) DR in soil equivalent soil mass for control and treatment plots from baseline values for two field experiments (changes from 2012 to 2020 for IP and 2018–2020 for DR). Significant p-values for treatments within year; $P \leq 0.05$ (*), $P \leq 0.01$ (**). Error bars represent standard error; IP ($n = 8$) and DR ($n = 6$).

to support greater BNF and plant production.

Overall, the increases in SOC stocks and N stocks in IP beyond the initial C and N inputs from the compost indicate that infrequent compost applications in irrigated cool-season pastures can increase SOC sequestration and total N stocks in the topsoil. Compost applications to pastures also increase forage yields providing a potential to increase revenues in addition to carbon markets or soil health incentive programs (Basche et al., 2020). However, the importance of SIC to total C below 20 cm merits careful consideration of how carbon sequestration is accounted for in semi-arid ecosystems. In soils with significant contributions of SIC to total C, the attention given to SOC sequestration under natural climate solution practices should expand to separately consider SIC sequestration impacts.

4.1. Organic amendments increased biomass production

We observed large increases in aboveground biomass after a second compost application in IP, but this response varied due to irrigation rates. In 2019, there were no differences between treatments when the irrigation schedule was irregular indicating that water availability is a major control on pasture productivity in this climate. While IP responded positively to the compost addition, the DR showed no response, which was likely a result of water limitation. Notably, neither site experienced an increase in annual invasive weeds over the experimental period with the compost (data not shown). Plant growth in arid and semi-arid climates is often limited by N availability (Hooper and Johnson, 1999), but other co-limiting resources like water and phosphorus

can impact plant responses (Chapin et al., 1987). Multiple, interacting resource limitations can inhibit growth of plant communities in grassland (Harpole et al., 2016) and shrubland ecosystems (James et al., 2005). In their global meta-analysis of grassland communities, DeMalach et al. (2017) found that simultaneous additions of nutrients and water led to larger biomass increases than either resource alone. The regular addition of water in IP in 2018 and 2020, the addition of multiple plant growth limiting macronutrients (i.e., N, P, K), and potential improvements in water holding capacity and water use efficiency, through the compost could explain the observed differences in aboveground biomass response between the two experiments.

While our estimates of standing root biomass were likely conservative due to our sampling method, standing root biomass tended to be higher in the compost treatment in IP from 0 to 20 cm relative to the unamended control during peak standing biomass. Ryals and Silver (2013) observed similar increases in root biomass under compost application in annual grasslands, particularly from 0 to 10 cm. A global meta-analysis of root responses to N additions suggests that in grasslands, root production is not negatively impacted by fertilization, while root turnover rate may actually increase (Peng et al., 2017). Increases in root turnover have important consequences for soil C accumulation. Root C, an important component of soil organic matter especially at depth (Rumpel and Kögel-Knabner, 2011) or where aboveground plant residues are not readily returned to the soil (Kramer and Gleixner, 2008), increases soil C via the proximity of roots to mineral surfaces, physio-chemical protection through soil aggregate formation (Dungait et al., 2012), and root exudation (Rasse et al., 2005).

4.2. Soil organic carbon and nitrogen stocks responded to infrequent amendment applications

The infrequent application of high-quality (low C:N) compost can increase SOC and total N stocks over short and medium durations (two to eight years) in irrigated, semi-arid agroecosystems. In an improved perennial pasture system in the Texas Southern High Plains, [Otuya et al. \(2020\)](#) observed increases in SOC and total N after one and a half years using composted animal manure applied at about half the rate (3.36 Mg ha^{-1}) as this study. In a three-year study in annual grasslands, [Ryals et al. \(2014\)](#) found that a single application of compost resulted in net C sequestration of 3 Mg C ha^{-1} . They also found that amendment C and N, and likely new plant C, were incorporated into physically protected soil organic matter pools. And, in their global meta-analysis, [Gravuer et al. \(2019\)](#) observed SOC concentration increases in 233 of 244 observations, and these increases occurred in two years or less in half of these observations but they do not necessarily indicate a net increase in soil C. Similarly, the application of biosolids in nearby semi-arid rangelands to this site increased total soil C and N in the top 8 cm of soil, particularly when applied at equal or greater rates than 21 Mg ha^{-1} , and increases in soil organic matter were still evident 13–14 years after initial application ([Ippolito et al., 2010](#)). However, in a similar irrigated pasture system in northern Colorado, [Mikha et al. \(2017\)](#) found that composted dairy manure applied at comparable rates as this study did not increase SOC or total N after two years. In this study, there were differences in SOC from 0 to 10 cm eight years after two compost applications in 2012 and 2018 in IP. However, these changes were much more pronounced after the second application measured in 2020.

Several mechanisms may explain the observed increases in SOC, and soil N in the topsoil in IP. First, increases in topsoil C and N, particularly from 0 to 10 cm, is likely directly related to the addition of the compost. We did not observe large, visible quantities of compost residue on the soil surface in the second and third experimental years, suggesting that it was readily decomposed and incorporated into the topsoil through downward movement by precipitation and irrigation events, and uptake and reallocation by plants. Soil microbial analyses undertaken at this site (data not shown) indicate that functional diversity in the top 10 cm of the soil shifted over the experimental period where microbial communities in the compost treatment were better able to process labile C inputs like glucose and xylose. Microbial products from decomposition of labile C inputs are critical for forming stable SOM, particularly mineral associated organic matter ([Haddix et al., 2016](#); [Lavalley et al., 2018](#)).

Because most of the aboveground inputs were removed from IP during haying operations, and the compost was not directly incorporated into the soil, root inputs, favorable soil mineralogy (i.e., high clay content of these soils), and possibly dissolved organic C were likely the important factors contributing to SOC accumulation in the topsoil. During the 2018 season, the compost had greater root production than the unamended control and greater overall plant productivity under the compost. The interactions of C inputs with the clay mineral fraction, in addition to the molecular composition of the C inputs, and the accessibility of this material to the microbial community ([Sollins et al., 1996](#); [Sokol et al., 2019](#)), all influence the transformation of plant and litter inputs into soil organic matter. Despite the increases in the topsoil, we did not observe a change over time in SOC stocks below 10 cm over the experimental period in either the compost treatment or the unamended control.

4.3. Soil inorganic carbon should not be overlooked

The quantification of net SOC sequestration rates in semi-arid and other regions with significant SIC stocks must consider how practices aimed at atmospheric CO_2 removal impact this important, and spatially variable, C pool in the short and long-term. Recommendations for measuring carbon sequestration for CO_2 removal focus exclusively on

SOC ([Billings et al., 2020](#); [Smith et al., 2020](#)), with little to no discussion of SIC. It is possible that management practices implemented to increase SOC sequestration decrease SIC stocks, which can result in a loss of CO_2 to the atmosphere in some instances depending on the dissolution processes acting on that inorganic C ([Monger et al., 2015](#)).

At our site, SIC contributed considerably to total soil C stocks in these soils, particularly in the subsurface ($> 20 \text{ cm}$). In IP, SIC stocks represented one-third of total soil C from 0 to 20 cm, but this increased to one-half in the 20–50 cm depth. In DR, the contributions of SIC to total soil C from 0 to 20 cm was like IP, but it represented nearly 60% of total soil C from 0 to 80 cm. We found a consistent negative correlation between SOC and SIC with depth across both sites (data not shown). These proportions are like other cultivated soils in the Central Great Plains ([Denef et al., 2008](#)). Land management, particularly irrigation and fertilization, can influence the SIC cycle with consequences for ecosystem C cycling ([Sanderman, 2012](#)). Irrigation of high pH soils can lead to soil bicarbonate formation through the stimulation of plant, root, and soil biotic activity ([Denef et al., 2008](#)). In contrast, frequent N fertilization and organic acid production from roots can decrease soil pH, acidifying the soil, and can lead to substantial decreases, or dissolution, of SIC over time ([Raza et al., 2020](#)). The spatial and temporal scales of these process must be taken into consideration when determining if pedogenic carbonate formation or dissolution processes are a net source, sink, or neutral carbon source to the atmosphere (for more see [Sanderman, 2012](#)).

We observed similar SIC stocks in the topsoil in both IP and DR suggesting that irrigation of these soils did not lead to an accumulation of SIC in the topsoil, consistent with other studies in the region ([Denef et al., 2008](#)). However, the high variability in SIC in IP and DR below 10 cm highlights the importance and challenges of carefully considering both the organic and inorganic C impacts of compost application to arid and semi-arid soils as well as the importance of measuring soil C stocks below the top 10 cm. While we did not observe differences between treatments in SIC losses under compost, the compost treatment in IP resulted in a loss of SIC over time since the first application in 2012 from 0 to 50 cm. The acidification of high pH soils through sustained synthetic N fertilization can decrease SIC stocks in some regions ([Wu et al., 2009](#); [Raza et al., 2020](#)). While compost application that increase SOC should help buffer pH changes, there is potential for acidification and SIC losses in naturally high pH soils common in semi-arid regions.

Historically subsoil C stock changes are not quantified in agricultural soils due to difficulty and costs ([Harrison et al., 2011](#)). In a long-term study of arable soils, [Gregory et al. \(2016\)](#) observed significant subsoil C losses nearly sixty years after land-use change indicating that deeper soil horizons are susceptible to C losses even when they are not directly impacted by land management decisions. And, in their study investigating the use of organic inputs to increase soil C in row crop systems, [Tautges et al. \(2019\)](#) observed significant losses of subsoil C under cover cropping, but important gains under poultry manure additions. In our case, we did not observe a change in SIC over the entire profile between treatments likely due to the high variability in measured SIC stocks. For example, in IP the standard error for SIC stocks was similar to SOC from 0 to 10 cm ($0.9 \text{ Mg OC ha}^{-1}$) while below 10 it was two times greater ([Table 1](#)). These differences were even greater for the DR site. While many studies quantifying soil C stock changes over time assume soils below about 20 cm can be considered stable ([Harrison et al., 2011](#)), this can only be ascertained through long-term evaluation of soil C pools deeper in the profile. It is not uncommon for variability in SOC to increase with depth ([Kravchenko and Robertson, 2011](#)), and this spatial variability may be even greater for SIC relative to SOC because it is influenced by multiple environmental controls (e.g., climate, parent material, age) ([Guo et al., 2006](#)).

To better understand changes in subsoil C, particularly SIC, our data suggests that a greater number of samples, smaller depth increments (e.g., 5 or 10 cm rather than 30 cm), and the addition soil fractionation analyses would reduce uncertainty and better inform on subsoil C

responses. *Ad hoc* power analysis can inform on the number of samples required to achieve a desired level of certainty in soil sampling designs, particularly for subsoil sampling schemes (Kravchenko and Robertson, 2011). Other protocols to quantify within-site variance can assist with modifications to sampling designs to determine the minimum number of samples required to detect a change (Arrouays et al., 2018). Using isotopic tracing techniques could also help determine if the decreases in SIC we observed were a net source of CO₂ to the atmosphere (Sanderman, 2012; Monger et al., 2015).

4.4. No differences in seasonal soil respiration

We could only measure soil respiration in the DR. The emissions trends generally followed the annual temperature and precipitation patterns during each experimental year (Fig. 1). The annual emissions trend might also be influenced by crop growth, but notably biomass production tended to increase over the experimental period in the degraded rangeland site while CO₂ emissions decreased over that same period. In our experiment, the application of compost did not increase seasonal cumulative soil respiration. These findings are contrary to previous experimental findings; a study in California annual grasslands observed 18–19% increases in soil respiration over three-years after compost application (Ryals and Silver, 2013) while a global meta-analysis of fifteen publications and 119 observations of soil respiration measured in the lab found that organic amendments increased soil respiration relative to unamended soils. The lack of treatment differences could be a result of low N application rates in DR; our application of compost in DR was approximately one-tenth of the application rate in Ryals and Silver (2013). Furthermore, the DR did not appear to be C or N limited given that there was no response of soil C:N to the amendment. Increases in soil respiration after organic amendment application are likely due to increases in root growth and root respiration, but also a result of the decomposition of labile organic inputs, particularly when C:N of the amendment is low, which favors fast growing soil microbial communities (Thangarajan et al., 2013). But, in our case, the C:N of the amendment and the soil were similar maintaining the stoichiometric balance of available organic substrates. It is also possible that we did not capture treatment differences in soil CO₂ flux immediately after precipitation events, despite efforts to sample within 24–72 h of these events. In semi-arid ecosystems, precipitation pulse events can stimulate large soil CO₂ fluxes, primarily attributable to heterotrophic respiration, which may be caused by the availability of accumulated, easily decomposed substrates (Inglis et al., 2009).

4.5. Quantifying the climate benefits of compost in semi-arid soils

The impacts of compost applications on soil N₂O and a complete life cycle assessment of feedstock diversion for organic amendment production merits further investigation. Agricultural soils are a significant contributor to global N₂O emissions (IPCC, 2019), and the application of excess N can stimulate N₂O emissions enough to offset soil C gains and perceived climate benefits (Zhou et al., 2017; Guenet et al., 2021). While amendment application did not increase soil N₂O in lab incubations (Ryals and Silver, 2013), the addition of crop residues and organic materials to agricultural soils can stimulate soil N₂O emissions via increased N supply and promoting anaerobic conditions needed for N₂O production (Chen et al., 2013). However, modest N additions that stay within the crop N requirements may not substantially impact N₂O emissions (Shcherbak et al., 2014), and composts along with other stable organic amendments are likely low risk for increased soil N₂O emissions (Charles et al., 2017). In our study, we observed an accumulation in total soil N beyond what was supplied from the compost suggesting no or minor N₂O losses. High frequency N₂O flux sampling (Parkin, 2008) supplemented with modeling (Giltrap et al., 2020) can help determine the soil N₂O impacts of amendment application in semi-arid systems. Finally, life cycle assessments of the alternative uses of amendment

feedstocks (e.g., manure, landfill, anaerobic digestion) can help determine the full environmental footprint of organic amendment applications (DeLange et al., 2013).

5. Conclusion

Moderate, infrequent applications of high-quality compost increased plant production and topsoil SOC stocks in an irrigated, cool season perennial pasture in northern Colorado but not an adjacent degraded rangeland. The differences in plant and soil responses to the compost across the two systems implies that water availability may be a limiting factor for increasing SOC stocks under this practice. The climate crisis and the growing competition for water between rural and urban users across the region will continue to favor irrigation well retirement and conversion of irrigated land use types to dryland agriculture. More research is needed to understand how shifts in irrigation and water availability will impact SOC stocks under this and other management practices implemented to build SOC stocks.

Our findings also raise important considerations for carbon and ecosystem services markets in this region, but also more broadly regarding the accounting of soil C stock changes under improved agricultural management practices. We observed variability in SIC at depth in both experiments, and a decrease over time in SIC under the compost in IP. A focus only on SOC stocks in the topsoil would overstate the benefit of this practice for building soil C in working lands as a natural climate solution. It is imperative that future research in semi-arid climates consider SIC and subsoil C dynamics to better ascertain how whole profile soil C and different soil C pools will respond before attempting to scale compost applications as a natural climate solution.

Code availability

Code is available upon request to the corresponding author.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data are available via Zenodo (DOI with raw data: [10.5281/zenodo.6412372](https://doi.org/10.5281/zenodo.6412372)).

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agee.2022.107969](https://doi.org/10.1016/j.agee.2022.107969).

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